Year 13 Further Maths - Roots of polynomials

Similar questions - Symmetric functions and substitution

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The quadratic equation $x^2 - 4x + 9 = 0$ has roots α and β.

(a) Write down $\alpha + \beta$ and $\alpha\beta$. **[1]**

(b) Hence find a quadratic equation with integer coefficients whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$. **[4]**

2 The roots of $2x^3 + 4x^2 - 3x + 5 = 0$ are α, β and γ.

(a) Find a cubic equation with integer coefficients whose roots are α^2, β^2 and γ^2. **[5]**

(b) Find the exact value of $\dfrac{\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2}{\alpha\beta\gamma}$. **[2]**

3 $\mathrm{f}(z) = z^3 - 3z^2 + kz - 5$ where k is a real constant. The roots of $\mathrm{f}(z) = 0$ are α, β, γ and $\alpha^2 + \beta^2 + \gamma^2 = -5$.

(a) Explain why $\mathrm{f}(z) = 0$ has only one real root. **[3]**

(b) Find the value of k. **[3]**

(c) Find a cubic equation with integer coefficients and roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$. **[2]**

4 The roots of $2x^3 - 5x^2 + 4x - 7 = 0$ are α, β, γ. Find a cubic equation with integer coefficients and roots $\alpha + \beta$, $\beta + \gamma$, $\gamma + \alpha$. **[6]**

Question 1

(a) $\alpha+\beta=4$, $\alpha\beta=9$

(b) Sum: $\alpha+\beta+\dfrac{\alpha+\beta}{\alpha\beta}=4+\dfrac{4}{9}=\dfrac{40}{9}$. Product: $\left(\alpha+\dfrac{1}{\beta}\right)\left(\beta+\dfrac{1}{\alpha}\right)=\alpha\beta+2+\dfrac{1}{\alpha\beta}=9+2+\dfrac{1}{9}=\dfrac{100}{9}$. Equation: $x^2-\dfrac{40}{9}x+\dfrac{100}{9}=0$, i.e. $9x^2-40x+100=0$

Question 2

(a) $\sum\alpha=-2$, $\sum\alpha\beta=-\frac{3}{2}$, $\alpha\beta\gamma=-\frac{5}{2}$. $\sum\alpha^2=4+3=7$. $\sum\alpha^2\beta^2=\left(-\frac{3}{2}\right)^2-2(-\frac{5}{2})(-2)=\frac{9}{4}-10=-\frac{31}{4}$. $\alpha^2\beta^2\gamma^2=\frac{25}{4}$. Equation: $x^3-7x^2-\frac{31}{4}x-\frac{25}{4}=0$, i.e. $4x^3-28x^2-31x-25=0$

(b) $\dfrac{-31/4}{-5/2}=\dfrac{31}{10}$

Question 3

(a) A cubic with real coefficients has at least one real root and non-real roots come in conjugate pairs. If all three roots were real then $\alpha^2+\beta^2+\gamma^2\geqslant 0$, but it equals -5. So there are two non-real roots and exactly one real root

(b) $\sum\alpha=3$, $\sum\alpha\beta=k$. $\sum\alpha^2=9-2k=-5$, so $k=7$

(c) Reverse the coefficients of z^3-3z^2+7z-5: $-5x^3+7x^2-3x+1=0$, i.e. $5x^3-7x^2+3x-1=0$

Question 4

$\sum\alpha=\frac{5}{2}$, so the new roots are $\frac{5}{2}-\gamma$, $\frac{5}{2}-\alpha$, $\frac{5}{2}-\beta$. Let $y=\frac{5}{2}-x$, so $x=\frac{5}{2}-y$: $2(\frac{5}{2}-y)^3-5(\frac{5}{2}-y)^2+4(\frac{5}{2}-y)-7=0$. Expanding gives $-2y^3+10y^2-\frac{33}{2}y+3=0$, i.e. $4y^3-20y^2+33y-6=0$. The required equation is $4x^3-20x^2+33x-6=0$